

Benchmarking Physics-Informed Neural Networks for Gravitational Potential Modeling

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ABSTRACT

Physics-Informed Neural Networks (PINNs) are widely studied as a mesh-free alternative to traditional partial differential equation solvers. However, their computational efficiency must be balanced against their ability to handle out-of-distribution data. This paper presents a benchmark comparing a PINN framework to a traditional adaptive Runge-Kutta numerical solver for the 3D radial Poisson equation. To isolate algorithmic speed from hardware acceleration, all evaluations were executed on a standard host CPU. The results show that the trained PINN achieves a **496.7x speedup** in forward simulation time over the numerical method in perturbed conditions, maintaining a stable latency of 0.00133 seconds. While the PINN achieves a high structural correlation ($r = 0.9793$) on the baseline system, it yields a high Mean Absolute Percentage Error (MAPE) of **936.05%** under high-frequency density perturbations. These findings suggest that while PINNs are highly efficient for nominal systems, their susceptibility to spectral bias limits their reliability in unmodeled, broken-symmetry fields.

Keywords: Physics-Informed Neural Networks (PINNs), Scientific Machine Learning, Numerical Methods, Computational Physics, Benchmarking

INTRODUCTION

Physics-Informed Neural Networks (PINNs) offer a way to solve partial differential equations by incorporating physical laws directly into the loss function during training. However, the balance between execution speed and predictive accuracy remains a primary challenge for practical deployment in high-throughput applications.

Traditional numerical integration methods are highly precise but demand significant computational resources, creating operational limits in real-time settings. This study evaluates whether PINNs can serve as a faster, scalable alternative for gravitational potential modeling. By evaluating the model against both analytical and numerical baselines, this work maps out the specific performance boundaries of PINNs under both ideal conditions and structural perturbations.

METHODS AND MATERIALS

0.1 The Governing Equation

This study models gravitational potential ϕ using the 3D radial Poisson equation:

$$\frac{d^2\phi}{dr^2} + \frac{2}{r} \frac{d\phi}{dr} = 4\pi G\rho(r)$$

where $G = 1.0$ is the gravitational constant and $\rho(r)$ is the mass density distribution. For the baseline configuration, the density follows a spherical Plummer profile:

$$\rho_{base}(r) = \frac{3M_{tot}}{4\pi a^3} \left(1 + \frac{r^2}{a^2}\right)^{-2.5}$$

where the total mass $M_{tot} = 1.0$ and the scale factor $a = 1.0$. To evaluate how the model handles broken spatial symmetries, an unmodeled high-frequency perturbation is added to the evaluation profile:

$$\rho_{perturbed}(r) = \rho_{base}(r) + 0.5 \sin(3r)$$

0.2 Baseline: Traditional and Analytical Methods

Two classical baselines were used for performance evaluation:

1. **Analytical Solution:** The unperturbed Plummer profile has a known closed-form solution: $\phi(r) = -1.0/\sqrt{r^2 + 1.0}$. This approach provides rapid evaluation but cannot produce a solution (*N/A*) once density perturbations are introduced.
2. **Numerical Integration:** An adaptive Runge-Kutta 4th/5th order solver (`solve_ivp`) was used to solve the equation sequentially across 500 random coordinates in the domain $r \in [0, 10]$, providing the baseline ground truth.

0.3 PINN Implementation

The PINN was built using the DeepXDE library with a PyTorch backend. The architecture consists of a fully connected neural network (FNN) with an input layer, 4 hidden layers of 64 neurons each, and a single output layer. Hyperbolic tangent (*tanh*) activation functions were used with Glorot normal weight initialization.

A Dirichlet boundary condition was enforced at the origin, setting $\phi(0) = -1.0$. The physics loss function minimized the residual of the unperturbed Poisson equation across 500 collocation points. The training process used a two-stage pipeline: an initial Adam optimization phase (10,000 epochs, learning rate $\eta = 0.001$) followed by L-BFGS refinement (2,000 iterations) to bring the total training loss to a stable threshold of 7.71×10^{-2} .

0.4 Computational Environment

To measure purely algorithmic speed and avoid hardware bias, GPU acceleration was disabled. All tasks were executed sequentially on an **Intel Xeon E3-1270 v3 CPU** (4 Cores, 8 Threads, 3.50 GHz) running CachyOS (Linux kernel 7.0.9).

RESULTS

The evaluation measured execution latency, Mean Absolute Error (MAE), Mean Absolute Percentage Error (MAPE), and the Pearson correlation coefficient (*r*) across 500 coordinates. Table 1 summarizes these metrics.

Table 1. Gravitational Potential Modeling Metrics on an Intel Xeon CPU

Method	System State	Latency (s)	Aligned MAE	MAPE (%)	Correlation (<i>r</i>)
Analytical	Perfect	0.00003	0.000030	0.01%	1.0000
Analytical	Perturbed	N/A	N/A	N/A	N/A
Numerical	Perfect	0.33212	0.000000	0.00%	1.0000
Numerical	Perturbed	0.66062	0.000000	0.00%	1.0000
PINN (Ours)	Perfect	0.00133	0.418358	195.19%	0.9793
PINN (Ours)	Perturbed	0.00133	0.647249	936.05%	0.2240

DISCUSSION

0.5 Algorithmic Speed Versus Traditional Solvers

The primary advantage of the PINN model is its execution speed. Traditional numerical solvers scale sequentially ($\mathcal{O}(N)$) because they must run independent integration steps for every target coordinate. On the test hardware, the numerical solver required **0.66062 seconds** to process the perturbed system.

In contrast, the trained PINN evaluates all 500 coordinates simultaneously through parallel matrix operations, resulting in a steady latency of **0.00133 seconds**. This represents a **496.7x speedup** over the numerical baseline. Additionally, the PINN latency remains completely constant regardless of whether the system is ideal or perturbed.

0.6 Generalization Limits and Spectral Bias

While the PINN provides clear speed advantages, the accuracy metrics reveal distinct functional limitations. In the ideal system, the network tracks the physical trend well, reaching a **0.9793 correlation coefficient**. The absolute error baseline (195.19% MAPE) indicates a uniform shift in the coordinate reference frame rather than an incorrect gradient shape.

However, when tested on the perturbed system, the model struggles to track the true function. The MAPE rises to **936.05%** while the structural correlation drops to **0.2240**. Because the network was optimized using a smooth, unperturbed Plummer profile, it exhibits known signs of spectral bias. It acts as a low-pass filter, completely smoothing out the high-frequency $0.5\sin(3r)$ density variations.

These results indicate that while vanilla PINN frameworks are useful for rapid inference in stable settings, they lack the adaptable accuracy of traditional numerical solvers when encountering unmodeled structural changes.